Current Eye Research

ISSN: 0271-3683 (Print) 1460-2202 (Online) Journal homepage: www.tandfonline.com/journals/icey20

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To cite this article: Yongli Xu, Run Sun, Man Hu & Hui Zeng (2024) A Dual-Modal Fusion Network Using Optical Coherence Tomography and Fundus Images in Detection of Glaucomatous Optic Neuropathy, Current Eye Research, 49:12, 1253-1259, DOI: 10.1080/02713683.2024.2375401

To link to this article: https://doi.org/10.1080/02713683.2024.2375401

Published online: 09 Jul 2024.

CURRENT EYE RESEARCH
2024, VOL. 49, NO. 12, 1253–1259
https://doi.org/10.1080/02713683.2024.2375401

A Dual-Modal Fusion Network Using Optical Coherence Tomography and Fundus Images in Detection of Glaucomatous Optic Neuropathy

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ABSTRACT

Purpose: We designed a dual-modal fusion network to detect glaucomatous optic neuropathy, which utilized both retinal nerve fiber layer thickness from optical coherence tomography reports and fundus images.

Methods: A total of 327 healthy subjects (410 eyes) and 87 glaucomatous optic neuropathy patients (113 eyes) were included. The retinal nerve fiber layer thickness from optical coherence tomography reports and fundus images were used as predictors in the dual-modal fusion network to diagnose glaucoma. The area under the receiver operation characteristic curve, accuracy, sensitivity, and specificity were measured to compare our method and other approaches.

Results: The accuracy of our dual-modal fusion network using both retinal nerve fiber layer thickness from optical coherence tomography reports and fundus images was 0.935 and we achieved a significant larger area under the receiver operation characteristic curve of our method with 0.968 (95% confidence interval, 0.937–0.999). For only using retinal nerve fiber layer thickness, we compared the area under the receiver operation characteristic curves between our network and other three approaches: 0.916 (95% confidence interval, 0.855, 0.977) with our optical coherence tomography Net; 0.841 (95% confidence interval, 0.749, 0.933) with Clock sectors division; 0.862 (95% confidence interval, 0.757, 0.968) with inferior, superior, nasal temporal sectors division and 0.886 (95% confidence interval, 0.815, 0.957) with optic disc sectors division. For only using fundus images, we compared the area under the receiver operation characteristic curves between our network and other two approaches: 0.867 (95% confidence interval: 0.781–0.952) with our Image Net; 0.774 (95% confidence interval: 0.670, 0.878) with ResNet50; 0.747 (95% confidence interval: 0.628, 0.866) with VGG16.

Conclusion: Our dual-modal fusion network utilizing both retinal nerve fiber layer thickness from optical coherence tomography reports and fundus images can diagnose glaucoma with a much better performance than the current approaches based on optical coherence tomography only or fundus images only.

ARTICLE HISTORY
Received 3 May 2024
Accepted 27 June 2024

KEYWORDS
Deep learning; fusion network; glaucoma; fundus image; optical coherence tomography

Introduction

Glaucoma refers to a collection of eye diseases that result in the impairment of the optic nerve, the pathway responsible for transmitting visual data from the retina to the brain. The deterioration of the optic nerve is frequently associated with elevated intraocular pressure, yet it is important to note that glaucoma can also occur in individuals with normal eye intraocular levels. Now glaucoma is the second leading cause of irreversible blindness in the world, and it is the main reason of visual disability across the world.[1,2] Many of glaucoma have no warning signs, and most patients may not notice a change in vision until the condition is in later stages.[3] We need a tool and method to help screening and diagnosis of glaucoma and it would be meaningful to the patients and doctors.

Optical coherence tomography (OCT) is a non-invasive and high-resolution technique and it has been used to analyze retinal tomographic structure parameters, including retinal nerve fiber layer (RNFL) thickness, choroidal thickness, and optic disc (OD) morphology. Structural changes in glaucoma can be detected with OCT, and a lot of evidence has shown that OCT is a valuable clinical tool for diagnosis of glaucoma.[4–8] The RNFL thickness from OCT has been demonstrated to provide accurate and detailed information of glaucoma diagnosis and progression.[9] Although OCT performs well in glaucoma diagnosis, it still has some misdiagnoses and missed diagnoses.[10–13] This is partially because the diagnosis just relies on average RNFL thickness or four quadrant thicknesses, rather than the full OCT scan image.[14]

A fundus image represents the projection of the fundus onto a 2D plane, captured by a monocular camera. In

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contrast to other eye scans like OCT images and angiographs, fundus images offer a non-invasive and cost-effective means of acquisition, rendering them ideal for large-scale screening purposes.[15] The utilization of fundus images for early eye disease screening holds significant clinical value. With deep learning demonstrating remarkable performance, it is more and more popular in glaucoma diagnosis.[16–19] Acharya et al.[19] introduced a screening approach utilizing different features extracted from Gabor transform applied to digital fundus images. Chen et al.[20] and Orlando et al.[21] applied classic deep learning algorithms to detect glaucoma from fundus images directly. Fu et al.[22] provided a deep learning approach with geometry transformation to gain additional image-relevant information, and improve the diagnosis performance.

The purpose of our study was to construct a deep learning algorithm to detect glaucoma, which can make use of both RNFL thickness from OCT and the full fundus image information.

Methods

Study design and participants

The study was conducted according to the tenets of the 2013 version of the Declaration of Helsinki and approved by the institutional review board of Beijing Tongren Hospital (identifier 1212017BJTR519). The medical ethics committee exempted the need for the informed consent of patients as this study was a retrospective review and analysis of fully anonymized color retinal fundus images and OCT reports.

Patients diagnosed with glaucoma were recruited consecutively from the outpatient clinics of the Ophthalmology Department at Beijing Tongren Hospital. Healthy volunteers, without any known eye conditions, were recruited from hospital staff, as well as the spouses and friends of patients. Each participant provided written informed consent, indicating their understanding of the OCT imaging procedure's role in diagnosing glaucoma.

In this study, two senior glaucoma specialists used the fundus images and OCT reports to detect glaucomatous optic neuropathy (GON). These two glaucoma specialists independently diagnosed each eye as GON and non-GON according to the criteria described in a previous study.[23] Eyes were diagnosed with glaucoma if any of the following conditions occurs: vertical cup-to-disc ratio (VCDR) ≥0.7, general rim thinning ≥60°, localized rim thinning <60° (11–1 o'clock or 5–7 o'clock), RNFL defects, or peripapillary hemorrhages. When an eye was judged as GON by at least one glaucoma specialist, that eye was judged as GON. In deep learning, larger datasets consistently yield better performance. According to the timeline of our research, we collected as much data as possible, and attempted to train our network. The sample size we used in the research yielded a great performance.

Data processing

In the data preprocessing, we processed fundus images and OCT reports separately. For the fundus image, data augmentation operations including random flipping, enhancing image brightness, and enhancing image contrast were performed. Enhancing image brightness and contrast is intended to enable the network to more accurately locate the areas that need to be focused on when extracting data features, thereby improving the efficiency of model training and helping the model fit better. Random flipping of images enhances the network's ability to distinguish between objects and non-objects from different angles, assisting the network in reducing issues such as missed recognition and misidentification due to different rotation angles of the original images during training.

For the OCT reports, we extracted the RNFL thickness curve, which represented the thickness of the RNFL within a certain radial distance around the OD, encompassing a full 360° circumference. This extracted curve was then recorded as a 360-dimensional RNFL vector. As depicted in Figure 1 (a), the x-axis of the RNFL thickness curve corresponds to the regions T-S-N-I-T in Figure 1 (b). The start of the x-axis corresponds to the middle of the region T in Figure 1(b). Figure 1(b) divides the OD into four sectors by inferior, superior, nasal temporal (ISNT) rule and it illustrates the varying regions and positions of the thickness curve coordinates on the fundus image. In our study, we divided the OD into 30 sectors and then we had 30-RNFL thickness. There were 360 elements in the 360-dimensional RNFL vector that we already extracted from the RNFL thickness curve, and we calculated the average for every 12 elements to obtain a new vector with dimension of 30. Each element of the 30-RNFL vector represented an average thickness with every 12° around the OD.

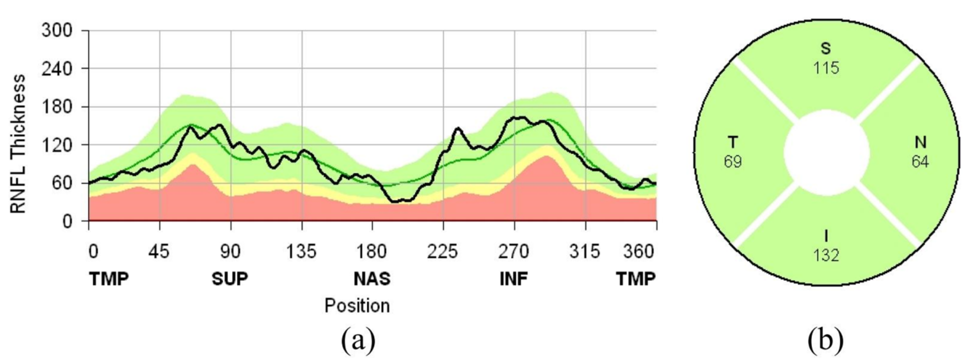

Figure 1. An example of retinal nerve fiber layer (RNFL) thickness curve (a) and quadrant thickness (b) in optical coherence tomography (OCT) report.

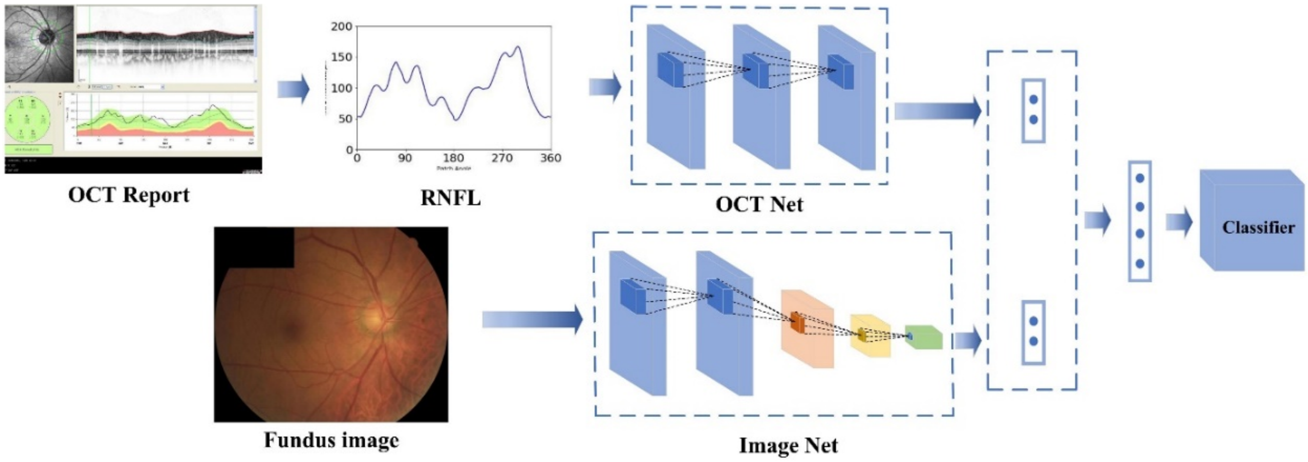

Figure 2. The detailed architecture of our dual-modal fusion network. OCT: optical coherence tomography; RNFL: retinal nerve fiber layer.

Design of our network for glaucoma classification

We developed a dual-modal fusion network, consisting of two stages: dual-modal feature extraction and feature fusion classifier. As it is shown in Figure 2, the first stage is the dual-modal feature extraction. The Image Net and OCT Net were employed to extract features from fundus images and RNFL thickness. In the Image Net, we used Inception v3 to extract features from fundus images, which is a Google-trained image classification model, and it optimizes the inception series by introducing sparse network structures. Through separable convolutions and combining multiple convolutional and pooling modules, it assembles the entire network structure at the module level, enhancing depth and width while reducing parameters, improving robustness. The overall structure of the network model remained unchanged, with only the last fully connected layer being modified to have two neurons, which were used to output a two-dimensional feature vector. In the OCT Net, we used an artificial neural network (ANN) with a three-layer perceptron (MLP) as the main framework for extracting features from 30-RNFL thickness. We designed a simple network comprising three fully connected layers: input layer, hidden layer, and output layer, with neuron counts of 30, 15, and 2. The activation function used was rectified linear unit (ReLU). After we extracted features in dual-modal feature extraction stage from the Image Net and OCT Net, we obtained two two-dimensional vectors representing different features from the fundus image and RNFL thickness.

In the feature fusion classifier stage, we concatenated these two features and performed feature fusion. We used fully connected layers as the final classifier, and then obtained the classification diagnosis results for glaucoma. The total loss was obtained through cross-entropy loss function, which is calculated based on the classification probability values output by the fusion network, without calculating separate loss functions for the Image Net and OCT Net. We used Adam Optimization Algorithm to improve training speed and reach convergence quickly with an initial learning rate of 3×10^{-4}, a batch size of 16, and a total of 300 epochs. Our network was implemented with PyTorch under Python.

Statistical analysis

We provided area under the receiver operating characteristic (AUC) curve for our dual-modal fusion network and some other algorithms utilizing fundus images only or RNFL thickness only. The 95% confidence intervals for AUC were calculated nonparametrically with bootstrap. We also used DeLong test to determine whether our dual-modal fusion network was statistically significantly different in AUC from those algorithms utilizing fundus images only or RNFL thickness only. We also provided accuracy, sensitivity, specificity, positive predictive value, and negative predictive value with 95% Clopper–Pearson intervals. All statistical analysis was performed using SPSS (Statistical Product and Service Solutions, Armonk, NY) and R (R Foundation for Statistical Computing, Vienna, Austria).

Results

Performance of our network compared with other algorithms

As it is shown in Table 1, there were 87 glaucoma patients (113 eyes), and 327 normal subjects (410 eyes) in total and we divided the data randomly into training set and testing set. In the 410 healthy eyes, there were 328 for training and 82 for testing. In the 113 eyes with glaucoma, there were 88 for training and 25 for testing. In the dual-modal fusion network, we chose Inception v3 in Image Net for fundus images and ANN for 30-RNFL thickness in OCT Net. It had the highest AUC, which was 0.968 (95% CI, 0.937–0.999) and its performance was better than those of only utilizing fundus images with Inception v3 or only utilizing 30-RNFL thickness with ANN, as it is shown in Figure 3.

As it is shown in Table 2, we provided AUC, accuracy, sensitivity, specificity, positive predictive value, and negative predictive value. According to the result of DeLong test, the AUC of our dual-modal fusion network was significantly higher than most of the algorithms with fundus images only or RNFL thickness only, especially compared to ResNet50 and VGG16 with fundus images only.

Table 1. Demographic characteristics of study subjects.

	No. of total individuals	No. of individuals without glaucoma	No. of individuals with glaucoma	No. of total eyes	No. of eyes without glaucoma	No. of eyes with glaucoma	Age, mean (SD)	No. of female/ total (%)
Training	307	245	62	416	328	88	47.3 ± 18.27	44
Testing	107	82	25	107	82	25	47.87 ± 16.73	43.93

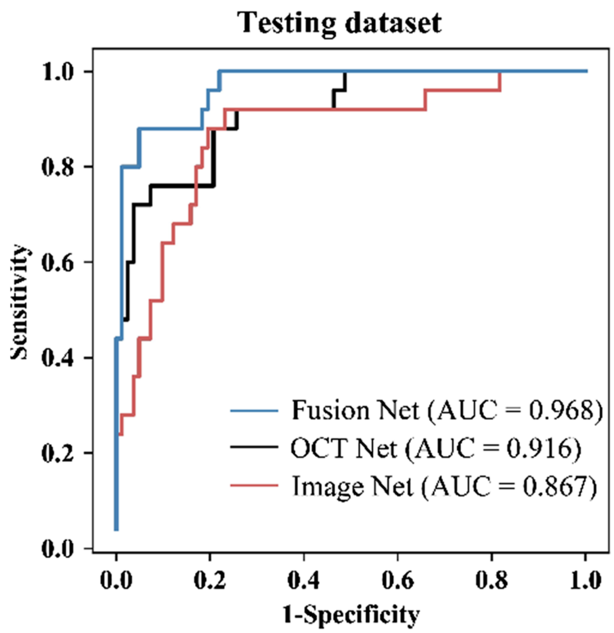

Figure 3. The receiver operating characteristic (ROC) curves of the dual-modal fusion network, OCT Net and Image Net. AUC: area under the receiver operation characteristic curve.

For only utilizing RNFL thickness, we compared ANN, which we chose for OCT Net with other algorithms with ISNT, Clock, and OD methods of dividing the OD. The OD shown in OCT reports follows the ISNT rule, which divides the circumference of the optic nerve fiber layer into four parts, where the S, I, T, and N regions are named upper, lower, nasal, and temporal, respectively. The angle of each region is 90° and the angle between the fan edge of each small region and the vertical or horizontal line is 45°. Method ISNT divides the OD followed the ISNT rule and then calculates the average thickness for each region. Method Clock and OD have four regions. Method Clock divides the OD horizontally and vertically. Method OD decreases the region N to 60° and increases the region T to 120° based on ISNT rule. We compared these three different methods with SVM (support vector machine) and our 30-RNFL thickness with ANN. As it is shown in Table 2, 30-RNFL thickness with ANN has a much better performance than the other three algorithms with 0.916 (95% CI, 0.855, 0.977) in AUC and 0.907 (95% CI, 0.850, 0.963) in accuracy.

We also compared some other algorithms only utilizing fundus images. ResNet50 and VGG16 are classic CNN models. As it is shown in Table 2, the inception v3, which we chose for Image Net had a much better performance than the other two algorithms with 0.867 (95% CI, 0.781, 0.952) in AUC and 0.832 (95% CI, 0.760, 0.964) in accuracy.

Interpretation of our network

We used two approaches to interpret our model: SHapley Additive exPlanations (SHAP)[24] and Gradient-Weighted Class Activation Mapping (Grad-CAM).[25] We employed a framework that utilizes SHAP to interpret the model performance with RNFL thickness, offering insights into the importance of 30 thicknesses for every prediction. These SHAP values represent the Shapley values of conditional expectation, illustrating the impact individually. We employed Grad-CAM to visualize significant regions highlighted by the model for prediction with fundus images. Grad-CAM utilizes gradient information from the final convolutional layer through back-propagation. These gradients are globally averaged to determine the importance of feature maps for the target. The resulting importance of the feature map serves as the model's confidence score.

In the SHAP method, SHAP values represent the importance of features. The larger the absolute value of the SHAP value, the higher the corresponding bar in the chart, indicating a greater positive or negative impact of that feature on the final result. In Figure 4, the SHAP values are associated with the RNFL curve based on the 30-RNFL thickness, which is the upper plot of Figure 4(a,c). The 30 bars in the lower plot of Figure 4(a,c) represent the SHAP value of the 30-RNFL thickness. Positive feature importance (red) indicates that the presence of this region would more likely lead the model to predict the sample as glaucomatous, while negative feature importance (blue) suggests that the presence of this region would more likely lead the model to predict the sample as normal eyes.

Figure 4(a) shows the performance on RNFL thickness that if we only use 30-RNFL thickness, and Figure 4(c) is for the dual-modal fusion network. Compared to the network only with 30-RNFL thickness, there are more regions utilized, as indicated by the larger number of red stripes with higher SHAP values in the fusion network. This suggested that these regions contributed to the model's correct judgments. Conversely, the overall lower SHAP values for the blue stripes imply that the negative impact of the blue regions on the model's predictions is smaller, leading to more accurate results. Furthermore, it is evident that both in the network with 30-RNFL thickness only and the fusion network, the regions corresponding to the red stripes in the RNFL were located within areas of significant changes in peaks, troughs, or slope of the curve. This indicates that the model pays close attention to the changes in thickness in these regions, which aligns with the reality that in glaucomatous pathology, there were significant changes in peaks, troughs, or slope compared to normal eyes.

When we only utilized fundus images to predict the glaucoma with inception v3, the key focus areas of the fundus

Table 2. Performance of our method and compared methods.

		AUC	p Value[a]	Accuracy	Sensitivity	Specificity	Positive predictive value	Negative predictive value
Retinal nerve fiber layer (RNFL) thickness	ANN	0.916 (0.855–0.977)	.145	0.907 (0.850–0.963)	0.720 (0.544–0.896)	0.963 (0.942–0.984)	0.857 (0.781–0.933)	0.919 (0.86–0.978)
	Clock	0.841 (0.749–0.933)	.015	0.822 (0.749–0.896)	0.600 (0.394–0.806)	0.890 (0.821–0.959)	0.761 (0.563–0.961)	0.895 (0.829–0.961)
	ISNT	0.862 (0.757–0.968)	.066	0.888 (0.827–0.949)	0.680 (0.483–0.877)	0.951 (0.904–0.999)	0.810 (0.626–0.993)	0.907 (0.844–0.97)
	OD	0.886 (0.815–0.957)	.048	0.869 (0.804–0.934)	0.640 (0.438–0.842)	0.939 (0.886–0.992)	0.625 (0.416–0.834)	0.879 (0.808–0.951)
Fundus image	Inception v3	0.867 (0.781–0.952)	.029	0.832 (0.760–0.964)	0.520 (0.310–0.730)	0.927 (0.869–0.984)	0.684 (0.454–0.914)	0.864 (0.791–0.937)
	ResNet50	0.774 (0.670–0.878)	.0004	0.813 (0.738–0.888)	0.400 (0.394–0.806)	0.939 (0.886–0.992)	0.667 (0.396–0.937)	0.834 (0.760–0.914)
	VGG16	0.747 (0.628–0.866)	.0007	0.822 (0.749–0.896)	0.48 (0.270–0.690)	0.926 (0.869–0.984)	0.667 (0.425–0.908)	0.853 (0.779–0.929)
Dual-modal fusion network	OCT: ANN Fundus image: Inception v3	0.968 (0.937–0.999)	–	0.935 (0.887–0.982)	0.76 (0.58–0.94)	0.988 (0.976–1)	0.950 (0.901–0.999)	0.930 (0.877–0.985)

AUC: area under the receiver operation characteristic curve; ANN: artificial neural network; ISNT: inferior, superior, nasal temporal; OD: optic disc.
Data are presented with 95% confidence interval.
[a]The DeLong test was used to calculate p values for comparison of areas under the receiver operator characteristic curve (AUC) between groups. Algorithms only utilizing RNFL thickness or only utilizing fundus images were compared with our fusion network.

images were around the macula and the edges of the images, as it is shown in Figure 4(b). However, in the dual-modal fusion network, the key focus areas of the fundus image were mainly located around the optic cup and OD, as it is shown in Figure 4(d). In clinical diagnosis of glaucoma, doctors mainly diagnose glaucoma by observing changes in the optic cup and OD, rather than the macula. Therefore, the heatmap proves the fusion network's model predicts the location of key focus areas more accurately.

Discussion

Recently, several reports of deep learning methods based on multimodal medical images for disease diagnosis and indicator prediction have been published.[26,27] In the field of intelligent glaucoma diagnosis, most studies are solely based on single-modal images, including fundus images, OCT, and visual fields, without fully exploring the correlations between multimodal images.[28–32] Wu et al.[33] provided a meta-analysis evaluating the performance of machine learning in detecting glaucoma with fundus and retinal OCT images based on different image types, different classifier types, and different datasets types. Compared to the results provided by Wu et al. our dual-modal fusion network had a higher AUC than the overall performance of machining learning algorithms in detecting glaucoma based on OCT. Also, compared to the results of neural network based on fundus and OCT separately, our dual-modal fusion network had a higher AUC.

In this study, we designed a deep neural network that integrated fundus images and OCT reports for glaucoma diagnosis. Compared with single-modal algorithms, the fusion network based on fundus images and OCT dataset pairs showed a higher AUC. Xiong et al.[34] proposed a deep learning network that combines OCT and visual field data, achieving higher accuracy than single-modal approaches. However, in clinical diagnosis, fundus photography is the most commonly used examination, and combining fundus images with other tests holds more practical clinical value.

In existing designs of multimodal deep neural networks, multimodal data are typically input to the deep neural network in the form of images.[26,27,34] In the fusion network designed in this paper, fundus photos were input as images, while OCT data were input as vectors. Here, the OCT vector represented local RNFL thickness values around the OD. We used two branch networks to extract features from fundus images and OCT vectors, then utilized two fully connected layers to fuse the features of both types, making a comprehensive diagnosis based on the fused features. This combination method provided a new paradigm for designing deep neural networks capable of integrating various forms of medical data. Integrating basic scalar information such as age, gender, family history, along with 24-hour intraocular pressure curves, with image information like fundus photos, OCT, and visual fields, will bring the diagnosis of glaucoma closer to the diagnostic thinking of ophthalmologists.

Despite the promising results, our study has several limitations. In this study, we only evaluated the algorithm on a dataset of 107 eyes collected from a single hospital. In the future, we will collect more data from multiple centers to conduct a more comprehensive evaluation of the algorithm. The purpose of our study was to automatically detect suspected glaucoma, as long as a specialist ruled for GON, we diagnosed this eye as suspicious GON. In future studies, we will consider asking a third, more senior expert to make a final diagnosis of the inconsistent eyes. For fundus images, we only utilized basic information in the fusion network without incorporating key indicators related to glaucoma-specific anatomical structures like the OD and cup, such as vertical cup-disc ratio. Regarding OCT data, we only utilized the average RNFL thicknesses of 30 local sectors, which may lead to the loss of detailed information in OCT images. Training deep neural networks using fundus images, OCT images, and crucial indicators inherent in both types of images is a direction for future exploration. While the heatmaps of fundus images and SHAP of RNFL thickness from OCT reflected the fusion network's focusing on key regions in both data forms, there is no explicit representation of the correlation of the same local lesion in the two types of data. Developing heatmap correlation techniques for multimodal data is a crucial means to enhance the interpretability of multimodal deep neural network models. Additionally, we only fused information from fundus images and OCT, neglecting other data forms such as visual fields and 24-hour intraocular pressure. Designing deep learning networks that integrate more modalities like fundus images, OCT, visual fields, and 24-hour intraocular pressure is an important research direction for the future.

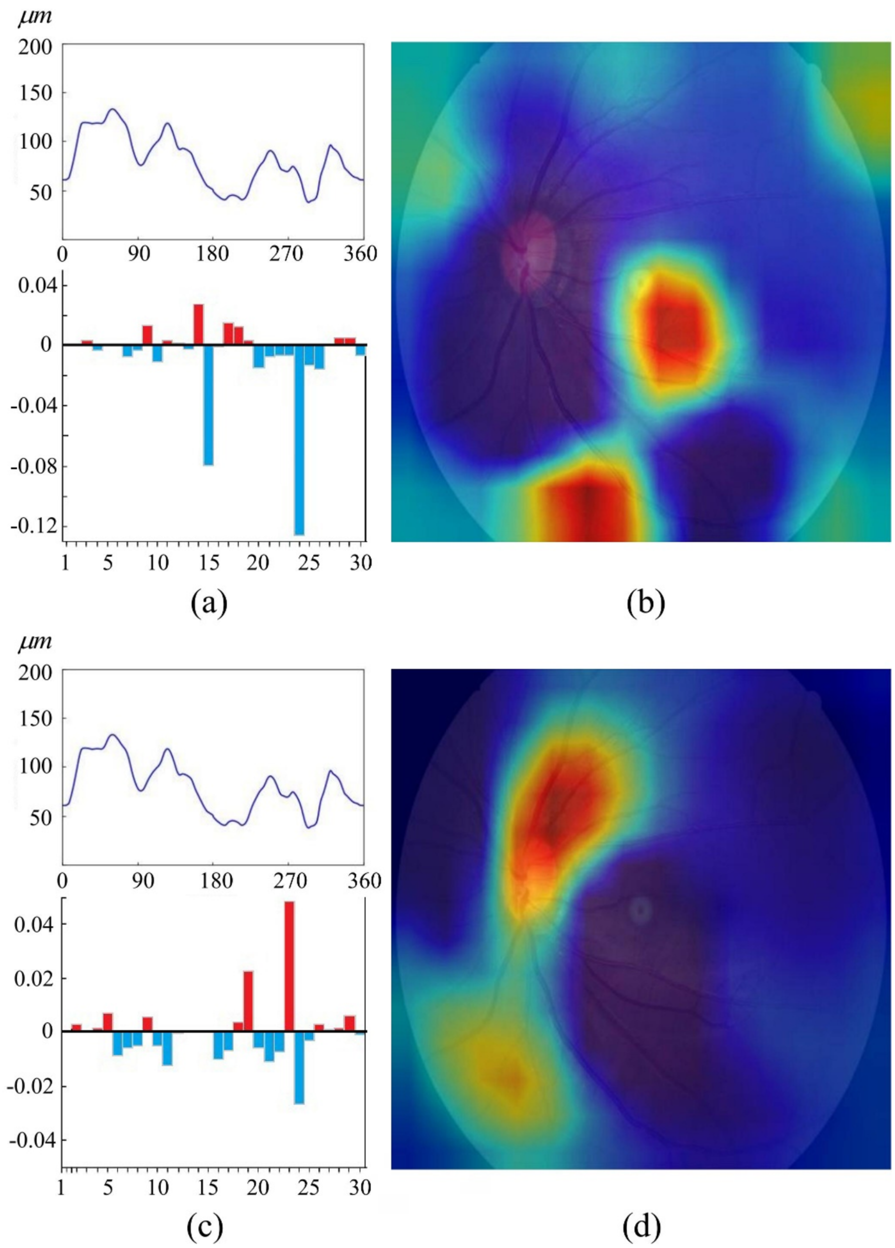

Figure 4. An example of model interpretation. (a) Model only utilizing retinal nerve fiber layer (RNFL) thickness; (b) model only utilizing fundus images; (c, d) the dual-modal fusion network utilizing both RNFL thickness and fundus images. The SHapley Additive exPlanations (SHAP) values for contributions of variables are shown in the histogram, which corresponds the RNFL curve. Red indicates that the variable may be positive factor for predicting glaucoma, whereas blue indicates a negative feature, as it is shown in (a) and (c). Heatmap generation with fundus photographs using Gradient-Weighted Class Activation Mapping (Grad-CAM), as it is shown in (b) and (d).

In conclusion, we designed a dual-modal fusion network to detect GON, which utilized both RNFL thickness from OCT reports and fundus images. Compared to the diagnosis on RNFL thickness from OCT only or fundus images only, our dual-modal fusion network had a much better performance.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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